THE DERIVED COMPLETION OF LOCAL COHOMOLOGY AS A DERIVED LIMIT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-dualizing/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The computation of local cohomology with support in a closed subscheme by the extended Cech complex. Invoked at the start of the counted proof (source lines 2344-2350). Provides the explicit complex whose derived completion is computed.
- The description of the derived completion of that complex as the complex of ordinary completions. Invoked inside the counted proof (source lines 2351-2358). Rewrites the derived completion termwise.
- The identification of a derived limit with the limit whenever the transition maps are surjective. Invoked inside the counted proof (source lines 2363-2364). Turns the termwise description into the asserted derived limit.
- The breaking of a derived limit into a short exact sequence of the first derived limit and the limit. Invoked inside the counted proof (source lines 2365-2366). Produces the displayed short exact sequences of part (2).

Source reference. Original lemma lemma-local-cohomology-adjoint: sources/stacks-dualizing/pinned-original-source.tex, source lines 1466–1466. Navigation pointer only: the counted text refers to the Stacks item lemma-local-cohomology-adjoint, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference more-algebra-lemma-break-long-exact-sequence-modules: sources/stacks-dualizing/pinned-original-source.tex, source lines 2366–2366. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-break-long-exact-sequence-modules, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own

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Source reference. Original reference more-algebra-lemma-compute-Rlim-modules: sources/stacks-dualizing/pinned-original-source.tex, source lines 2364–2364. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-compute-Rlim-modules, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference more-algebra-lemma-when-derived-completion-is-completion: sources/stacks-dualizing/pinned-original-source.tex, source lines 2352–2352. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-when-derived-completion-is-completion, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let I and J be ideals in a Noetherian ring A. Let M be a finite A-module. Set $Z = V(J)$. Consider the derived I-adic completion $R\Gamma_Z(M)^\wedge$ of local cohomology. Then*

(1) *we have $R\Gamma_Z(M)^\wedge = R\lim R\Gamma_Z(M/I^nM)$, and*
(2) *there are short exact sequences*

$$0 \to R^1 \lim H^{i-1}_Z(M/I^nM) \to H^i(R\Gamma_Z(M)^\wedge) \to \lim H^i_Z(M/I^nM) \to 0$$

In particular $R\Gamma_Z(M)^\wedge$ has vanishing cohomology in negative degrees.

Proof. Suppose that $J = (g_1, \ldots, g_m)$. Then $R\Gamma_Z(M)$ is computed by the complex

$$M \to \prod M_{g_{j_0}} \to \prod M_{g_{j_0}g_{j_1}} \to \ldots \to M_{g_1g_2\ldots g_m}$$

by Lemma lemma-local-cohomology-adjoint. By More on Algebra, Lemma more-algebra-lemma-when-derived-completion-is-completion the derived I-adic completion of this complex is given by the complex

$$\lim M/I^nM \to \prod \lim (M/I^nM)_{g_{j_0}} \to \ldots \to \lim (M/I^nM)_{g_1g_2\ldots g_m}$$

of usual completions. Since $R\Gamma_Z(M/I^nM)$ is computed by the complex $M/I^nM \to \prod (M/I^nM)_{g_{j_0}} \to \ldots \to (M/I^nM)_{g_1g_2\ldots g_m}$ and since the transition maps between these complexes are surjective, we conclude that (1) holds by More on Algebra, Lemma more-algebra-lemma-compute-Rlim-modules. Part (2) then follows from More on Algebra, Lemma more-algebra-lemma-break-long-exact-sequence-modules. □